Kubernetes Migration Analysis: DigitalOcean → AWS EKS

Prepared: 2026-02-20 **Scope:** Namespaces `services` and `prex` **Source cluster:** DigitalOcean Kubernetes (DOKS) **Target:** Amazon Elastic Kubernetes Service (EKS)

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1. Cluster Overview

Property	Value
Kubernetes version	v1.32.10
Container runtime	containerd 1.6.33
Node OS	Debian GNU/Linux 12 (Bookworm)
Total nodes	79
Total namespaces	~170 (many tenant namespaces)
Ingress controller	ingress-nginx (two deployed versions: v1 legacy + v2 active)

1.1 Node Pools

Pool Name	Instance Type	DO Spec	Recommended AWS Equivalent	Node Count
`standard-pool`	s-6vcpu-16gb	6 vCPU / 16 GB RAM	`m6i.xlarge` (4 vCPU / 16 GB) or `m6i.2xlarge` (8 vCPU / 32 GB)	40
`delayed-job`	s-6vcpu-16gb	6 vCPU / 16 GB RAM	`m6i.xlarge` or `m6i.2xlarge`	38
`monitoring`	s-2vcpu-4gb	2 vCPU / 4 GB RAM	`t3.medium` (2 vCPU / 4 GB)	1

> ***Note:*** *AWS `m6i.xlarge` provides 4 vCPU / 16 GB for general workloads. If current CPU headroom on DO nodes is needed, `m6i.2xlarge` (8 vCPU / 32 GB) is the safe choice. The consulting team should validate against actual `kubectl top nodes` usage data. The `delayed-job` pool name implies workload-specific scheduling — node taints and labels will need to be reproduced on EKS managed node groups.*

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2. Namespace: `services`

2.1 Deployments

Name	Replicas	Container(s)	Image Registry	CPU Req	Mem Req	CPU Limit	Mem Limit
`bear-services`	1	`services`, `services-hds`	`registry.emrbear.com` ⚠️	250m + 250m	1Gi + 1Gi	none	none

`bear-services-nginx`	1	`nginx`	`nginx:alpine` (Docker Hub)	250m	200Mi	none	none
`bear-x12`	2	`app`	`registry.emrbear.com` ⚠️	20m	200Mi	none	none
`bear-xero`	1	`bear-xero`	`registry.emrbear.com` ⚠️	250m	250Mi	none	none
`docker-registry`	1	`registry`	`registry:2.7.1` (Docker Hub)	400m	5Gi	none	none
`trip-form-filler`	3	`app`	`ghcr.io/emrbear`	50m	700Mi	none	none

> ***Observation:*** *No CPU limits are set on any workload in this namespace. This is a best-practice gap that should be addressed in the migration.*

2.2 StatefulSets / DaemonSets / CronJobs / Jobs

None found in `services` namespace.

2.3 Services

Name	Type	Port(s)	Selector
`bear-services`	ClusterIP	80	`bear-services-nginx`
`bear-services-backend`	ClusterIP	8080, 8081	`bear-services`
`bear-x12`	ClusterIP	3000	`bear-x12`
`bear-xero`	ClusterIP	4567	`bear-xero`
`docker-registry`	ClusterIP	5000	`docker-registry`
`latex-service`	ClusterIP	9000	`latex-service` (no active deployment found)
`nginx-test`	ClusterIP	80	`nginx` test pod
`trip-form-filler`	ClusterIP	4567	`trip-form-filler`

All services are ClusterIP — external traffic enters via Ingress only.

> ***`latex-service`:*** *Has a ClusterIP service but no corresponding deployment was found. Likely orphaned — confirm before migration.*

2.4 Ingress Resources

Name	Host	Backend Service	Ingress Class
`bear-services`	`services.emrbear.com`	`bear-services:80`	`nginx`
`bear-x12`	`parser.emrbear.com`	`bear-x12:80` (via nginx)	`nginx`

`bear-xero-ingress`	`bear-xero.emrbear.com`	`bear-xero:80` (via nginx)	`nginx`
`docker-registry`	`registry.emrbear.com`	`docker-registry:5000`	`nginx`

> ***TLS:*** *No TLS termination configured at the Ingress level for these hosts. Verify whether TLS is handled upstream (CDN/proxy) or if this needs to be added on EKS.*

2.5 Persistent Volume Claims

None. No stateful storage in this namespace.

2.6 Horizontal Pod Autoscalers

None configured.

2.7 ConfigMaps

Name	Purpose
`kube-root-ca.crt`	Auto-injected cluster CA
`registry-config`	Docker registry configuration
`services-nginx-config`	Nginx configuration for bear-services
`xero-config`	Xero API connector configuration

2.8 Secrets ⚠️ Migration Blockers

Secret Name	Type	Notes
`bear-x12-secrets`	Opaque	Application credentials for X12 parser
`bear-xero-secrets`	Opaque	Xero OAuth credentials
`emrbear-registry-secret`	`kubernetes.io/dockerconfigjson`	Pull credentials for `registry.emrbear.com` — **critical: tied to in-cluster registry**
`github-registry`	`kubernetes.io/dockerconfigjson`	Pull credentials for `ghcr.io`
`registry-secret`	Opaque	Docker registry auth/storage secret
`services-db-secrets`	Opaque	Database connection credentials

> ***Action required:*** *All Opaque secrets must be audited, re-provisioned, and stored in AWS Secrets Manager or SSM Parameter Store. Consider using the AWS Secrets Manager CSI driver or External Secrets Operator to inject them into EKS pods without storing them as plain Kubernetes Secrets.*

2.9 Service Accounts

Default service account only. No custom RBAC.

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3. Namespace: `prex`

3.1 Deployments

Name	Replicas	Container(s)	Image	CPU Req	Mem Req
`cerner-web`	2	`web`	`ghcr.io/emrbear/prex-cerner`	**none**	**none**
`idology-web`	1	`web`	`ghcr.io/emrbear/prex-idology`	**none**	**none**
`production-idology-web`	1	`web`	`ghcr.io/emrbear/prex-idology`	**none**	**none**
`production-surescripts-web`	2	`production-web`	`ghcr.io/emrbear/prex-surescripts`	**none**	**none**
`production-surescripts-worker`	1	`production-worker`	`ghcr.io/emrbear/prex-surescripts`	**none**	**none**
`saaspass-web`	1	`web`	`ghcr.io/emrbear/prex-saaspass`	**none**	**none**
`staging-idology-web`	1	`web`	`ghcr.io/emrbear/prex-idology`	**none**	**none**
`staging-surescripts-web`	1	`staging-web`	`ghcr.io/emrbear/prex-surescripts`	**none**	**none**
`staging-surescripts-worker`	1	`staging-worker`	`ghcr.io/emrbear/prex-surescripts`	**none**	**none**

> ***Critical gap:*** *Zero CPU or memory requests/limits are defined for any workload in the* `prex` *namespace. This means Kubernetes cannot make informed scheduling decisions and EKS node sizing will be guesswork.* ***Resource profiling (e.g., via Kubecost or VPA in recommendation mode) must be completed before EKS node group sizing.***

3.2 CronJobs

Name	Schedule	Notes
`production-surescripts-directory-full-sync`	`0 0 * * 0` (weekly, Sundays)	Surescripts full directory sync
`production-surescripts-directory-nightly-sync`	`0 0 * * 1-5` (Mon–Fri nightly)	Surescripts nightly sync
`staging-surescripts-directory-full-sync`	`0 3 * * 6` (weekly, Saturdays)	Staging full sync
`staging-surescripts-directory-nightly-sync`	`0 3 * * 1-5` (Mon–Fri nightly)	Staging nightly sync

3.3 Failed Jobs ⚠️

Three Jobs are in **Failed** state with extended durations, indicating pre-existing operational issues:

Job Name	Status	Age
`production-surescripts-directory-full-sync-29397600`	Failed	89 days
`production-surescripts-directory-nightly-sync-29373120`	Failed	106 days
`staging-surescripts-directory-nightly-sync-28818900`	Failed	491 days

> *These should be investigated and resolved independently of the migration. Their presence may indicate connectivity or credential issues with the Surescripts directory service.*

3.4 Services

Name	Type	Port	Notes
`cerner-web`	ClusterIP	80	Active
`idology-web`	ClusterIP	80	Active (staging)
`production-idology-web`	ClusterIP	80	Active (production)
`production-surescripts-web`	ClusterIP	80	Active
`saaspass-web`	ClusterIP	80	Active
`staging-idology-web`	ClusterIP	80	Active
`staging-surescripts-web`	ClusterIP	80	Active
`surescripts-web`	ClusterIP	80	**No matching deployment — likely orphaned**

3.5 Ingress Resources

Name	Host	Ingress Class
`surescripts-production-ingress`	`surescripts-production.emrbear.com`	`nginx`
`surescripts-staging-ingress`	`surescripts-staging.emrbear.com`	`nginx`

> *Cerner, Idology, and Saaspass services have no Ingress defined — they may be accessed via internal cluster routing only, or via the main ingress in another namespace. Confirm routing before migration.*

3.6 Persistent Volume Claims

None. No stateful storage in this namespace.

3.7 Horizontal Pod Autoscalers

None configured.

3.8 ConfigMaps

Name	Notes
`cerner-config`	Cerner integration configuration
`idology-config`	IDology identity verification config
`prex-elasticsearch-config`	⚠️ Elasticsearch endpoint config — dependency to identify
`prex-proxy-config`	⚠️ Proxy configuration — may reference DO-specific endpoints
`saaspass-config`	SaasPass SSO configuration
`surescripts-config`	Surescripts production config
`surescripts-config-staging`	Surescripts staging config

3.9 Secrets ⚠️ Migration Blockers

Secret Name	Type	Notes
`cerner-secrets`	Opaque	Cerner API credentials
`github-docker-registry`	`kubernetes.io/dockerconfigjson`	Pull credentials for `ghcr.io`
`idology-secrets`	Opaque	IDology API credentials
`prex-aws-secrets`	Opaque	**Already contains AWS credentials** — this namespace has an existing AWS integration
`prex-postgres-secrets`	Opaque	PostgreSQL connection credentials (production)
`prex-postgres-staging-secrets`	Opaque	PostgreSQL connection credentials (staging)
`saaspass-secrets`	Opaque	SaasPass credentials
`surescripts-connection-cert`	Opaque	**TLS client certificate for Surescripts connectivity** — requires careful migration
`surescripts-secrets`	Opaque	Surescripts API credentials (production)
`surescripts-staging-secrets`	Opaque	Surescripts API credentials (staging)

3.10 Service Accounts

Default service account only. No custom RBAC.

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4. Ingress Controller

Property	Current (DOKS)	Target (EKS)

Controller	ingress-nginx (v2 active, v1 legacy)	ingress-nginx (same) or AWS Load Balancer Controller
Service type	`NodePort` (10.245.201.84)	`LoadBalancer` → AWS NLB or ALB
TLS termination	Not configured at Ingress level	Add cert-manager + ACM integration

> *Two ingress controller deployments exist (`nginx-ingress-controller-manual` on NodePort 32086/31015, and `nginx-ingress-v2` on NodePort 31184/32662). The v2 controller is the active one. The legacy v1 installation should be decommissioned. On EKS, the recommended approach is to expose ingress-nginx via an AWS Network Load Balancer or use the AWS Load Balancer Controller for native ALB Ingress.*

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5. Storage

5.1 Storage Classes (Cluster-Wide)

Name	Provisioner	Reclaim Policy	Volume Binding	Expandable
`do-block-storage` (default)	`dobs.csi.digitalocean.com`	Delete	Immediate	Yes
`do-block-storage-retain`	`dobs.csi.digitalocean.com`	Retain	Immediate	Yes
`do-block-storage-xfs`	`dobs.csi.digitalocean.com`	Delete	Immediate	Yes
`do-block-storage-xfs-retain`	`dobs.csi.digitalocean.com`	Retain	Immediate	Yes

> ***All storage classes are DO-specific.*** *They must be replaced with EKS equivalents using `ebs.csi.aws.com`:*
>
> | *DO Storage Class* | *EKS Replacement* |
> |---|---|
> | *`do-block-storage`* | *`gp3` (default EKS StorageClass)* |
> | *`do-block-storage-retain`* | *Custom `gp3` StorageClass with `reclaimPolicy: Retain`* |
> | *`do-block-storage-xfs`* | *Custom `gp3` StorageClass with `fsType: xfs`* |
> | *`do-block-storage-xfs-retain`* | *Custom `gp3` StorageClass with `fsType: xfs` + `reclaimPolicy: Retain`* |

5.2 PVCs in Scope

No PersistentVolumeClaims exist in the `services` or `prex` namespaces. No stateful storage migration is required for these two namespaces specifically. Other namespaces in the cluster (not in scope here) may have PVCs.

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6. Container Image Registries

Registry	Used By	Migration Action

`registry.emrbear.com` (self-hosted, in-cluster)	`bear-services`, `bear-x12`, `bear-xero`	**High priority:** This registry runs as a pod in the `services` namespace. It must be migrated to AWS ECR (or kept external) before these workloads can be redeployed. Images must be pushed to ECR and all image references updated.
`ghcr.io/emrbear/*`	`trip-form-filler`, all `prex` workloads	No migration required — GHCR is cloud-agnostic. Pull secrets need to be re-provisioned.
`nginx:alpine`, `registry:2.7.1`	`bear-services-nginx`, `docker-registry`	Public Docker Hub images — no migration needed.

> *The in-cluster Docker registry (`docker-registry` deployment, exposed at `registry.emrbear.com`) is a **critical dependency and migration blocker**. It stores images used by other services in the cluster. Migration sequence must ensure this registry is replaced or mirrored to ECR before any dependent workloads are redeployed.*

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7. External Dependencies Identified

Dependency	Evidence	Action Required
PostgreSQL database	`prex-postgres-secrets`, `prex-postgres-staging-secrets`	Identify host (likely DO Managed Database). Plan migration to AWS RDS PostgreSQL.
Elasticsearch	`prex-elasticsearch-config` ConfigMap	Identify host (self-hosted or DO Managed). Plan migration to AWS OpenSearch or self-hosted on EKS.
Surescripts directory service	CronJobs, `surescripts-connection-cert`	External SaaS — cluster needs outbound connectivity to Surescripts endpoints. TLS client certificate must be re-provisioned in EKS.
Xero API	`bear-xero` deployment, `xero-config`	External SaaS — ensure outbound IP allowlisting is updated if Xero has IP restrictions.
IDology	`idology-web` deployments	External SaaS — same IP allowlisting concern.
Cerner	`cerner-web` deployment	External SaaS — same IP allowlisting concern.
Existing AWS integration	`prex-aws-secrets` in `prex` namespace	Some AWS service is already being called from this namespace. Identify which service and ensure IAM role/credentials are migrated correctly (prefer IRSA on EKS over static credentials).

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8. Migration Risk Summary

8.1 High Priority (Blockers)

#	Issue	Namespace	Impact
1	**In-cluster Docker registry** hosts images for active workloads	`services`	Cannot redeploy `bear-services`, `bear-x12`, `bear-xero` without migrating images to ECR first
2	**All Secrets must be migrated** — 16 Opaque secrets across both namespaces	Both	No workload will function without credentials
3	**Surescripts TLS client certificate** (`surescripts-connection-cert`)	`prex`	Must be re-issued or carefully transferred; likely tied to a specific endpoint/IP
4	**PostgreSQL database location** unknown	`prex`	If on DO Managed Databases, data migration to RDS must precede app migration
5	**Storage class replacement** — all DO CSI classes are incompatible with EKS	Cluster-wide	Equivalent EKS StorageClasses must be pre-provisioned

8.2 Medium Priority

#	Issue	Namespace	Impact
6	**No resource requests/limits on `prex` workloads**	`prex`	Cannot right-size EKS node groups; risk of OOMKill or CPU starvation
7	**No CPU limits on `services` workloads**	`services`	Same concern — limits should be added
8	**Ingress controller dual-version** (v1 legacy still deployed)	`nginx-ingress`	Decommission v1 before migration to avoid confusion
9	**Three long-running failed CronJobs**	`prex`	Pre-existing issue; should be resolved before migration to avoid inheriting broken state
10	**Outbound IP change** for Xero, IDology, Cerner, Surescripts	Both	EKS nodes will have different public IPs — notify external partners if IP allowlisting is in place
11	**`prex-aws-secrets` uses static AWS credentials**	`prex`	Should be replaced with EKS IRSA (IAM Roles for Service Accounts)

8.3 Low Priority / Housekeeping

#	Issue	Notes
12	Orphaned `latex-service` ClusterIP in `services`	No deployment backing it — confirm and remove

13	Orphaned `surescripts-web` ClusterIP in `prex`	No deployment backing it — confirm and remove
14	No HPAs in either namespace	Consider adding after migration with actual AWS metrics
15	`prex-proxy-config` may reference DO-specific proxy endpoints	Audit content before migration

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9. Recommended Migration Approach

1. **Pre-migration (NOW):**
   - Add resource requests/limits to all `prex` workloads and missing limits in `services`
   - Resolve the three failed Surescripts CronJobs
   - Audit `prex-proxy-config` for DO-specific endpoints
   - Identify and confirm PostgreSQL and Elasticsearch locations
   - Contact external partners (Xero, IDology, Cerner, Surescripts) about IP changes
2. **Registry migration:**
   - Create AWS ECR repositories for `bear-services`, `bear-services-hds`, `bear_x12`, `bear-xero`
   - Push all current images from `registry.emrbear.com` to ECR
   - Update deployment manifests to reference ECR
   - Migrate `docker-registry` deployment to ECR (or decommission if ECR fully replaces it)
3. **EKS cluster setup:**
   - Kubernetes version: v1.32 (matches current DOKS version — no version gap)
   - Create three managed node groups: `standard-pool`, `delayed-job`, `monitoring`
   - Reproduce node pool labels/taints for workload affinity
   - Install ingress-nginx v2 with AWS NLB (LoadBalancer service type)
   - Install AWS EBS CSI driver and create equivalent StorageClasses
   - Install cert-manager + ACM integration for TLS
4. **Secrets migration:**
   - Provision all secrets into AWS Secrets Manager
   - Deploy External Secrets Operator to sync into Kubernetes Secrets
   - Replace `prex-aws-secrets` static credentials with IRSA
5. **Workload migration (namespace by namespace):**
   - `services` namespace first (lower external dependency complexity)
   - `prex` namespace second (more external dependencies, failed jobs to resolve)
6. **DNS cutover:**
   - Update DNS for `services.emrbear.com`, `parser.emrbear.com`, `bear-xero.emrbear.com`, `registry.emrbear.com`, `surescripts-production.emrbear.com`, `surescripts-staging.emrbear.com`

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10. Workload Summary

Namespace	Deployments	CronJobs	Failed Jobs	Services	Ingresses	PVCs	Secrets
`services`	6 (16 total pods)	0	0	8	4	0	6
`prex`	9 (13 total pods)	4	3	8	2	0	10
Total	**15**	**4**	**3**	**16**	**6**	**0**	**16**

Report generated from live cluster data via `kubectl` on 2026-02-20.